

melody via speaker LS1. This melody sounds when the DC pump motor starts watering the plants.

IC 7809 is a fixed linear voltage regulator that maintains constant 9V supply to the circuit.

Construction and testing

An actual-size PCB layout for the automatic plants watering system with melody is shown in Fig. 3 and its components layout in Fig. 4. After assembling the circuit on the PCB, connect 12V batteries across BATT.1 and BATT. 2.

Fix LDR1 at a suitable location so that sunlight falls directly on it. Adjust VR1 for appropriate output. Select DIP1 switch in appropriate position for desired watering duration. Adjust VR2 for fine time adjustment. Rechargeable batteries are preferred for smooth working of the system. The proposed enclosure of the automatic plants watering system is shown in Fig. 5. **EFY**

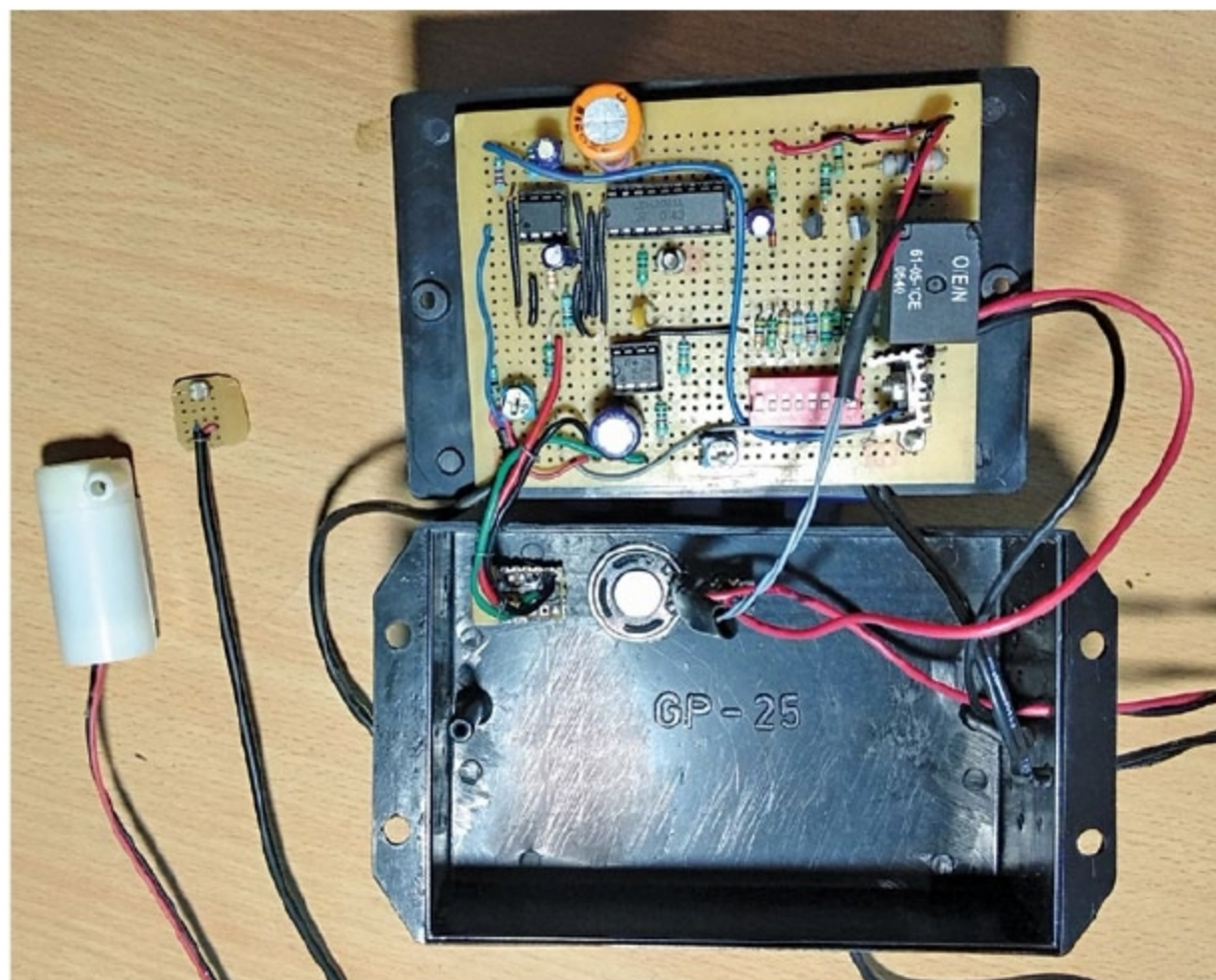


Fig. 5: Proposed enclosure for automatic plants watering system

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